

GEOGRAPHY

PHYSICAL GEOGRAPHY



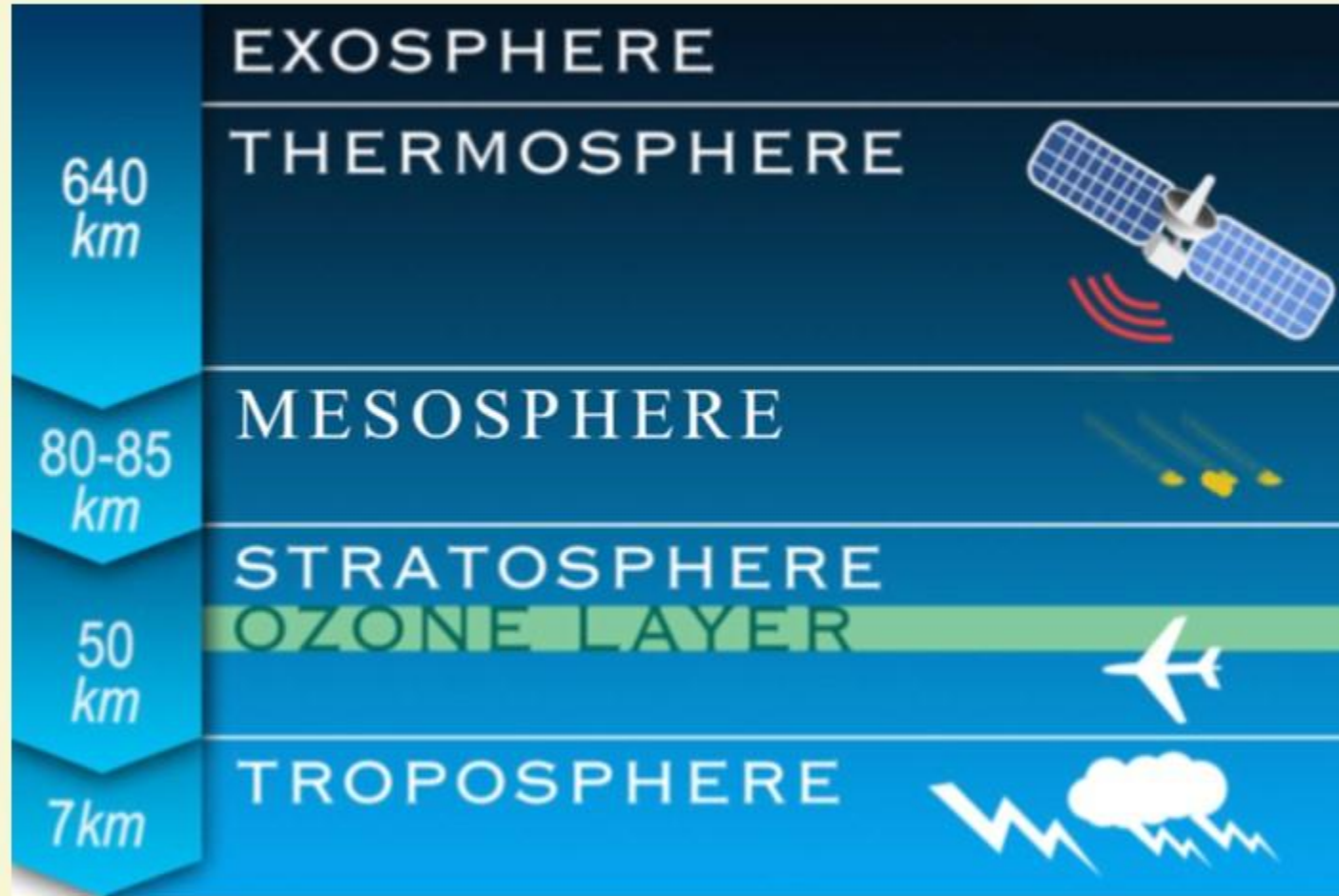
CLIMATOLOGY

MODULE – 5

STRUCTURE OF ATMOSPHERE

- **TROPOSPHERE**

STRUCTURE OF THE ATMOSPHERE

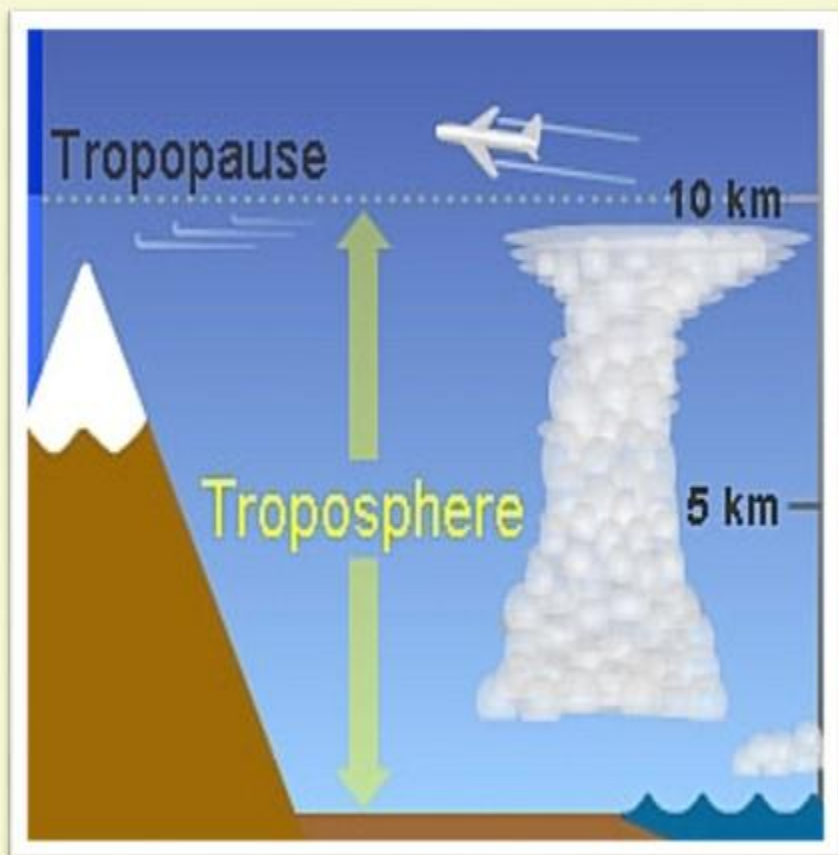


The atmosphere is composed of the following layers:

1. Troposphere
2. Stratosphere
3. Mesosphere
4. Thermosphere
5. Exosphere

STRUCTURE OF THE ATMOSPHERE

1.TROPOSPHERE

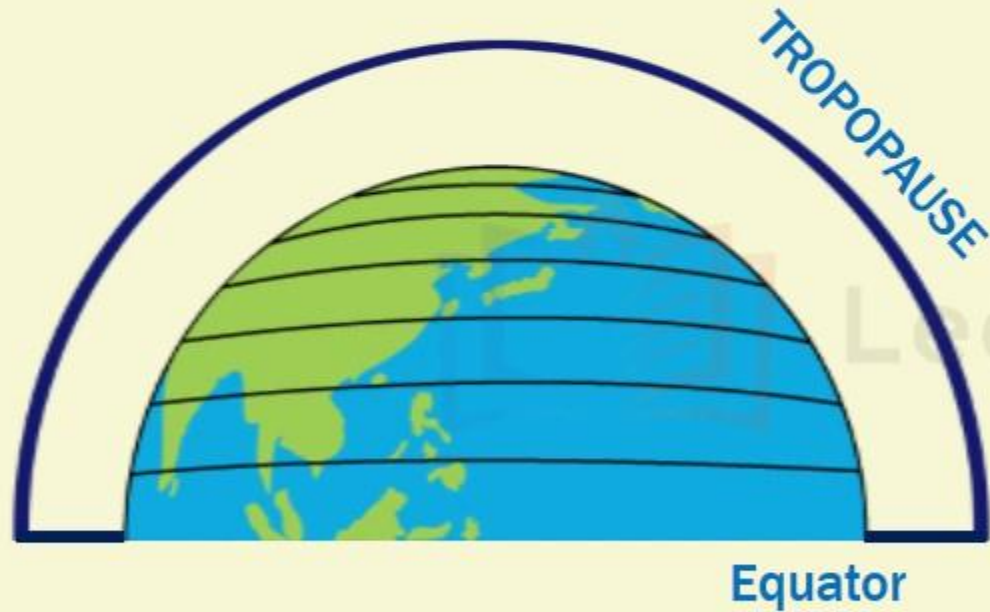


- 1** This is the abode of all living organisms
- 2** Almost all weather phenomena occur here
- 3** Its average height is around 13km. The height is roughly around 6-8km around the poles and 16-18km around the equator.
- 4** Upper limit of the troposphere is called as the tropopause which means 'where the mixing stops'.

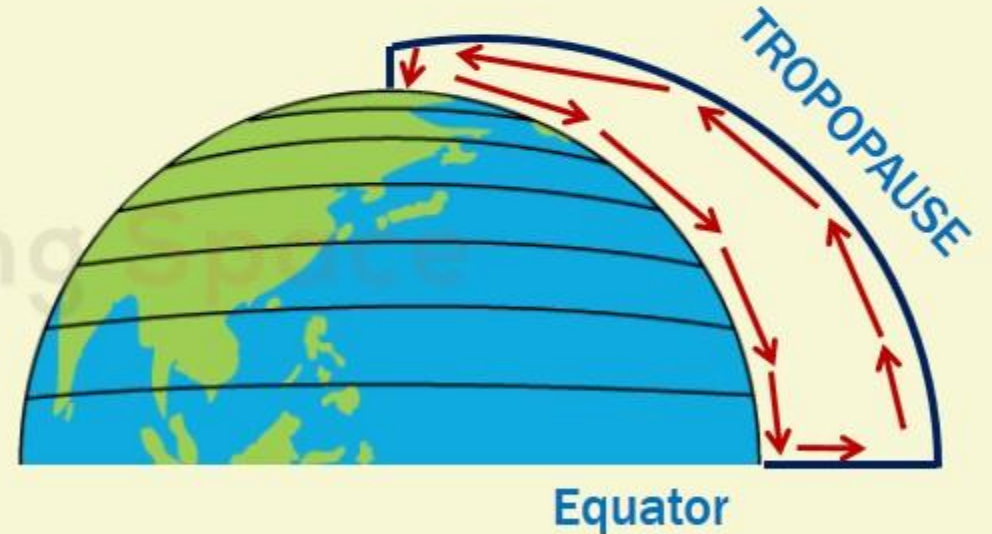
STRUCTURE OF THE ATMOSPHERE

1.TROPOSPHERE

EXPECTATION



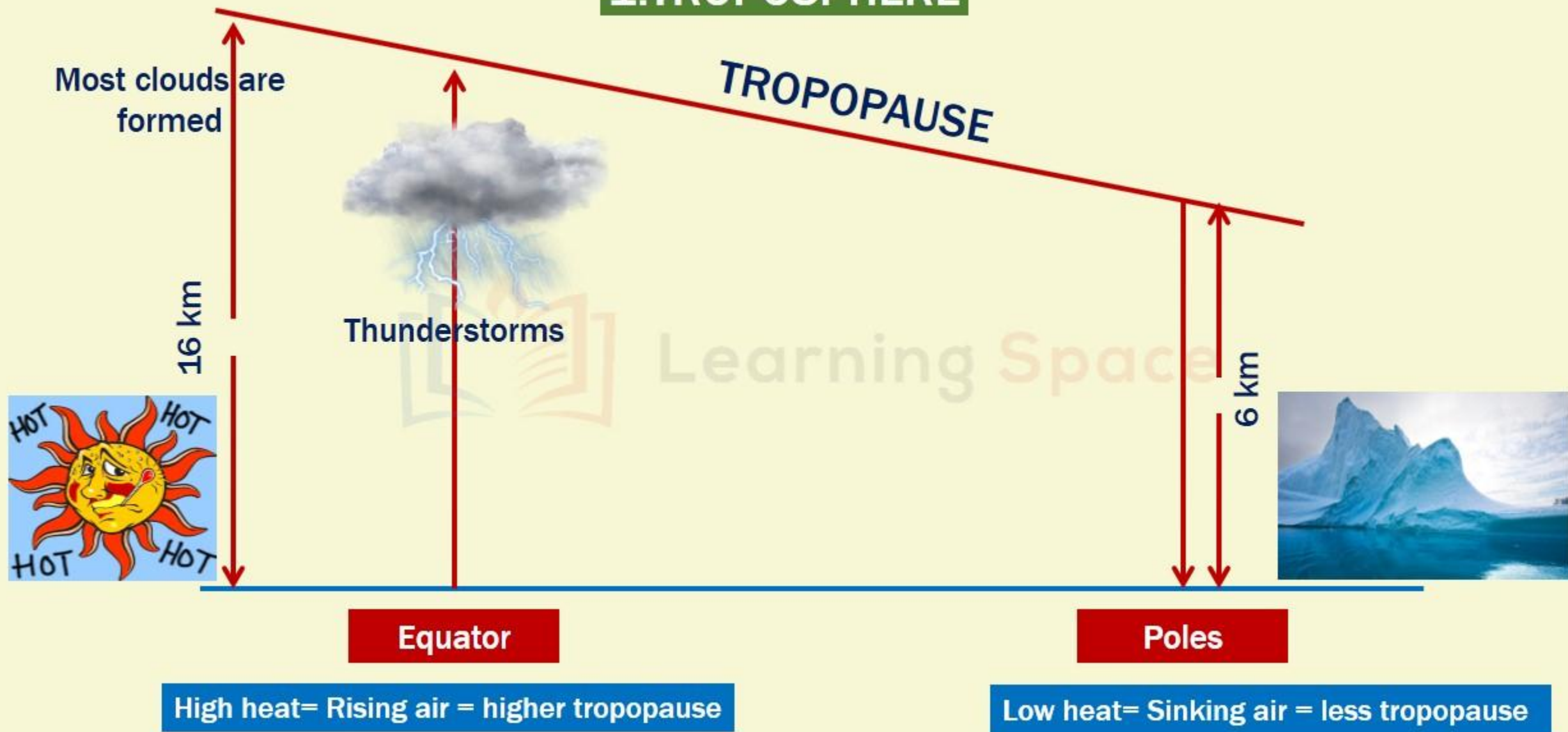
REALITY



Wider(16km) at the equator and
narrow(6km) at the poles

STRUCTURE OF THE ATMOSPHERE

1.TROPOSPHERE



STRUCTURE OF THE ATMOSPHERE

1.TROPOSPHERE

Why is the height of tropopause maximum around the equator and minimum around the poles?

REASON 1

At the Equator:

- 1 High heat in the equatorial areas will heat up air mass in those areas.
- 2 The heated up air parcel will expand, become less dense and consecutively rises up.
- 3 Thus a strong convection current is produced because of excessive heating and this will push the tropopause upward.

At the Poles:

- 1 The reverse happens in the case of poles.
- 2 Air gets very cold, becomes dense and starts sinking.
- 3 This causes the tropopause to remain at a lower height as well.

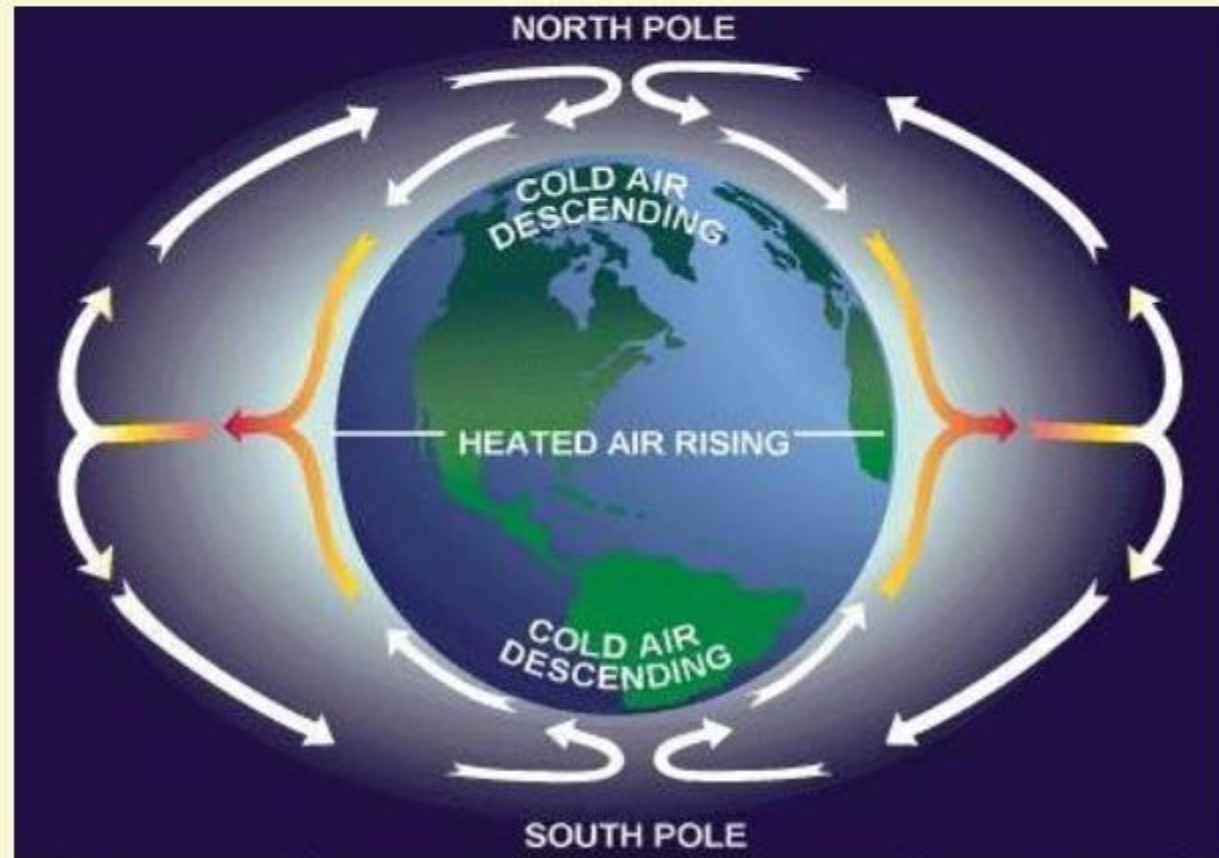
STRUCTURE OF THE ATMOSPHERE

1.TROPOSPHERE

Why is the height of tropopause maximum around the equator and minimum around the poles?

REASON 1

Heated air in equator is rising while cold air is descending in the poles



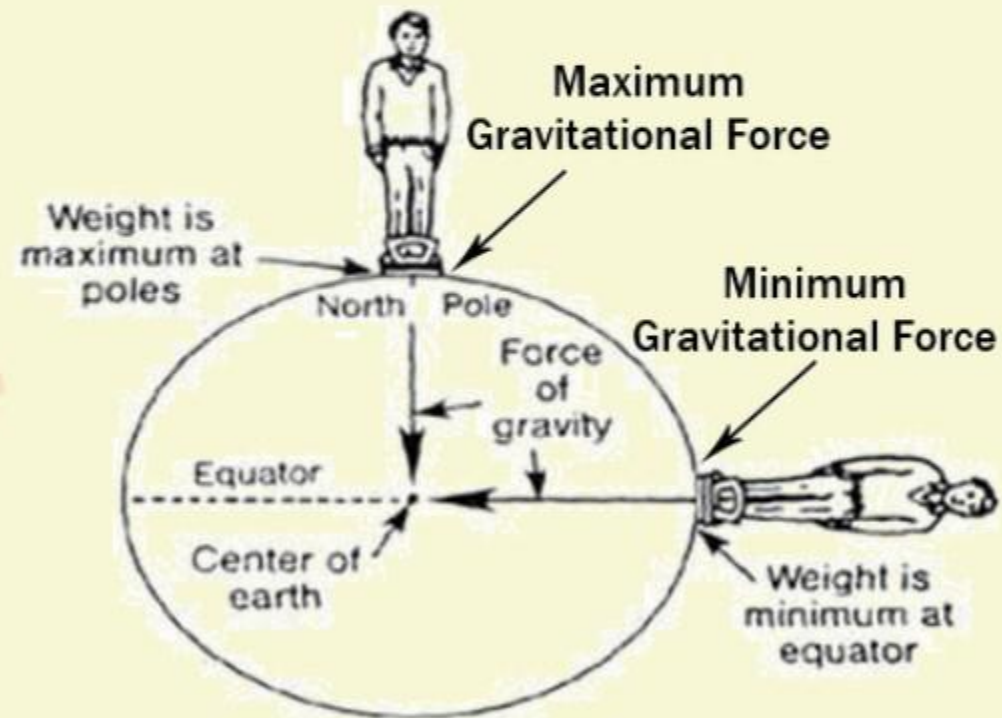
STRUCTURE OF THE ATMOSPHERE

1.TROPOSPHERE

Why is the height of tropopause maximum around the equator and minimum around the poles?

REASON 2

- 1 Gravity increases from equator to poles as the earth is not a perfect sphere (oblate spheroid). That means the gravitational force is more over poles.
- 2 Hence the atmosphere is pulled with more force near the poles than at the equator leading to contraction of the atmosphere at the poles and expansion at the equator.



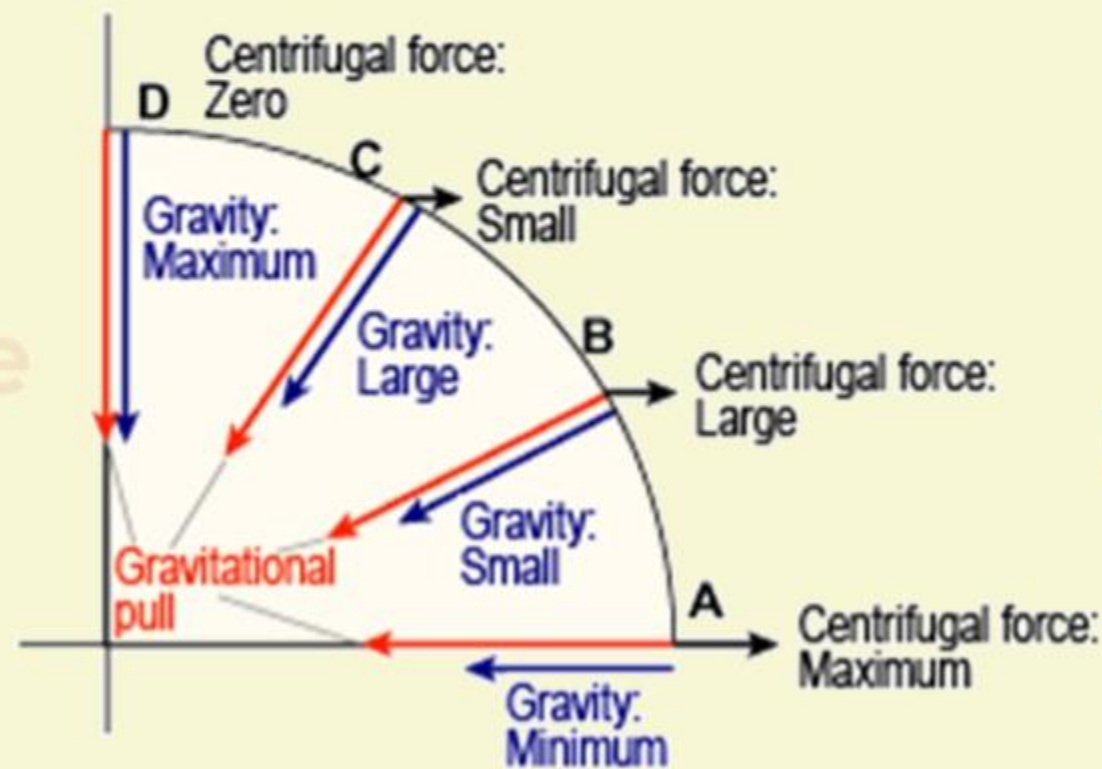
STRUCTURE OF THE ATMOSPHERE

1.TROPOSPHERE

Why is the height of tropopause maximum around the equator and minimum around the poles?

REASON 3

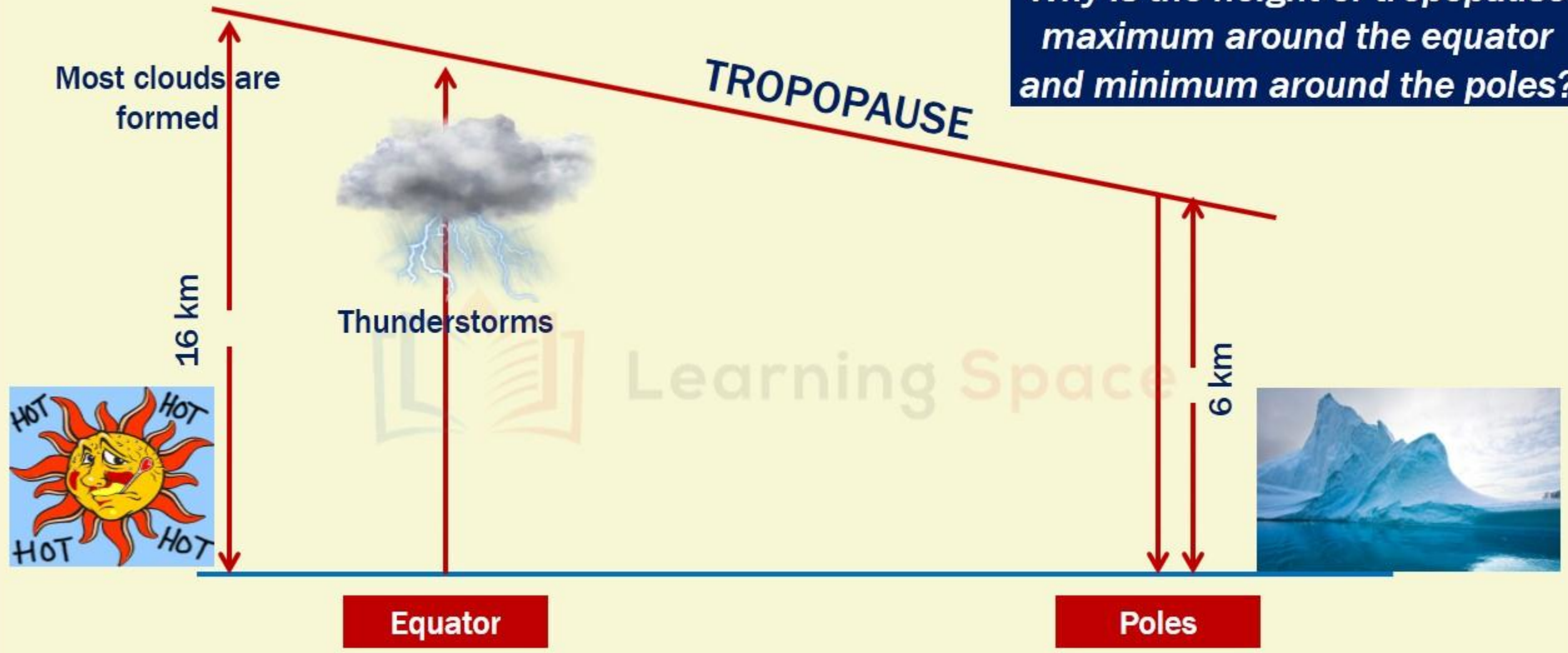
- 1** The centrifugal force (the force acting on a body moving in a circular path which is directed away from itself) due to Earth's rotation is maximum at Equator because the speed of the rotating earth is greatest at the equator.
- 2** Thus the atmosphere tends to bulge out at the equator because of this force.



STRUCTURE OF THE ATMOSPHERE

1.TROPOSPHERE

Why is the height of tropopause maximum around the equator and minimum around the poles?



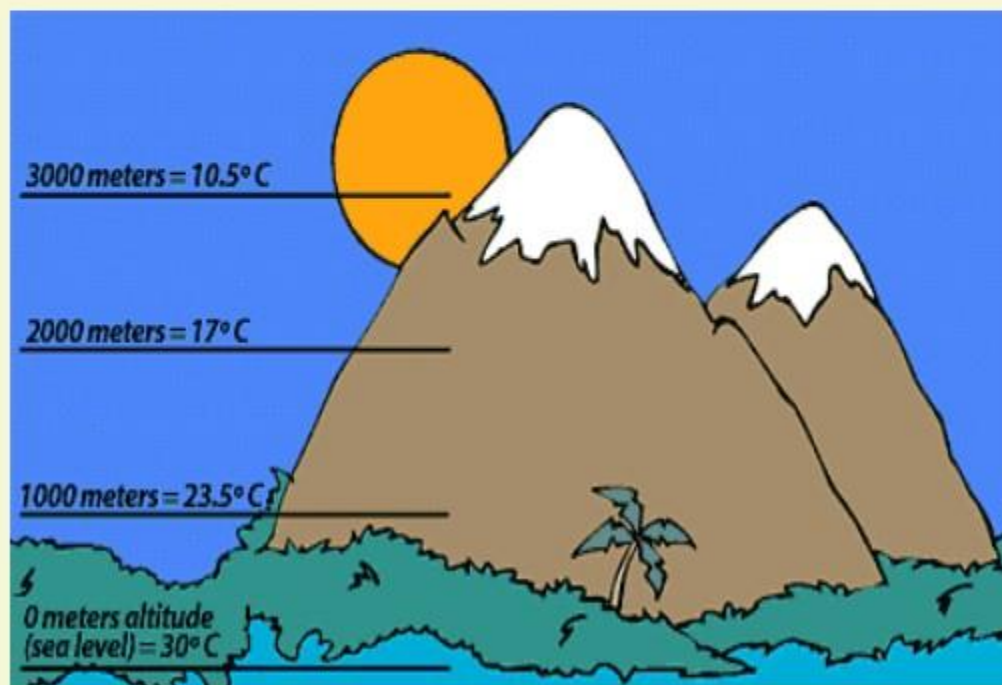
High heat= Rising air = higher tropopause

Low heat= Sinking air = less tropopause

STRUCTURE OF THE ATMOSPHERE

1.TROPOSPHERE

- 5** Temperature drops with increasing altitude at the rate of 6.5 degree Celsius per 1000m. This rate is known as **Normal Lapse Rate(NLR)**.

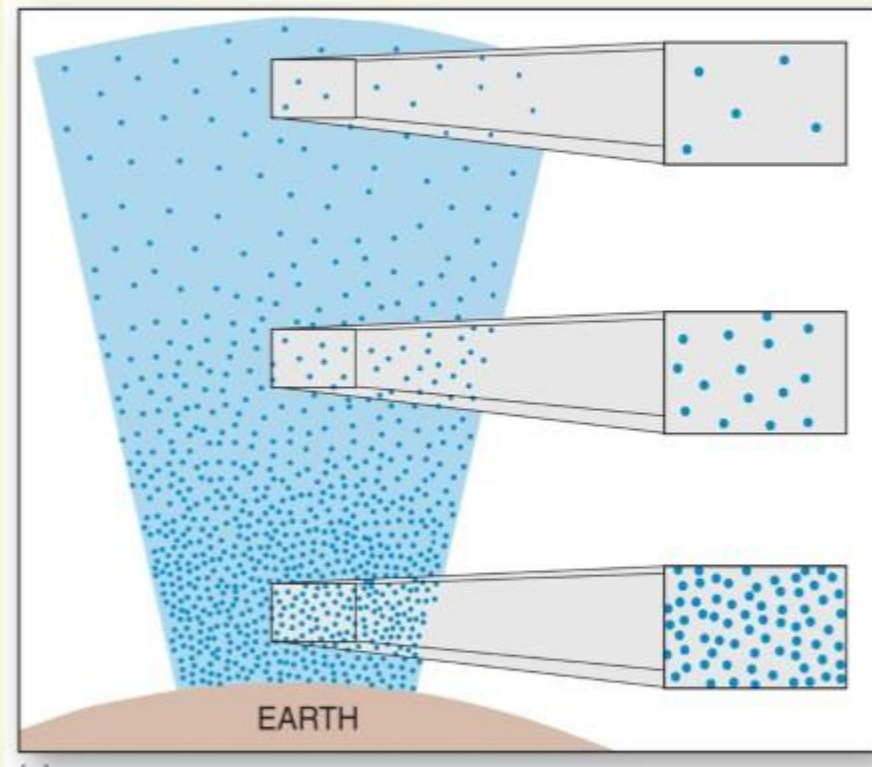
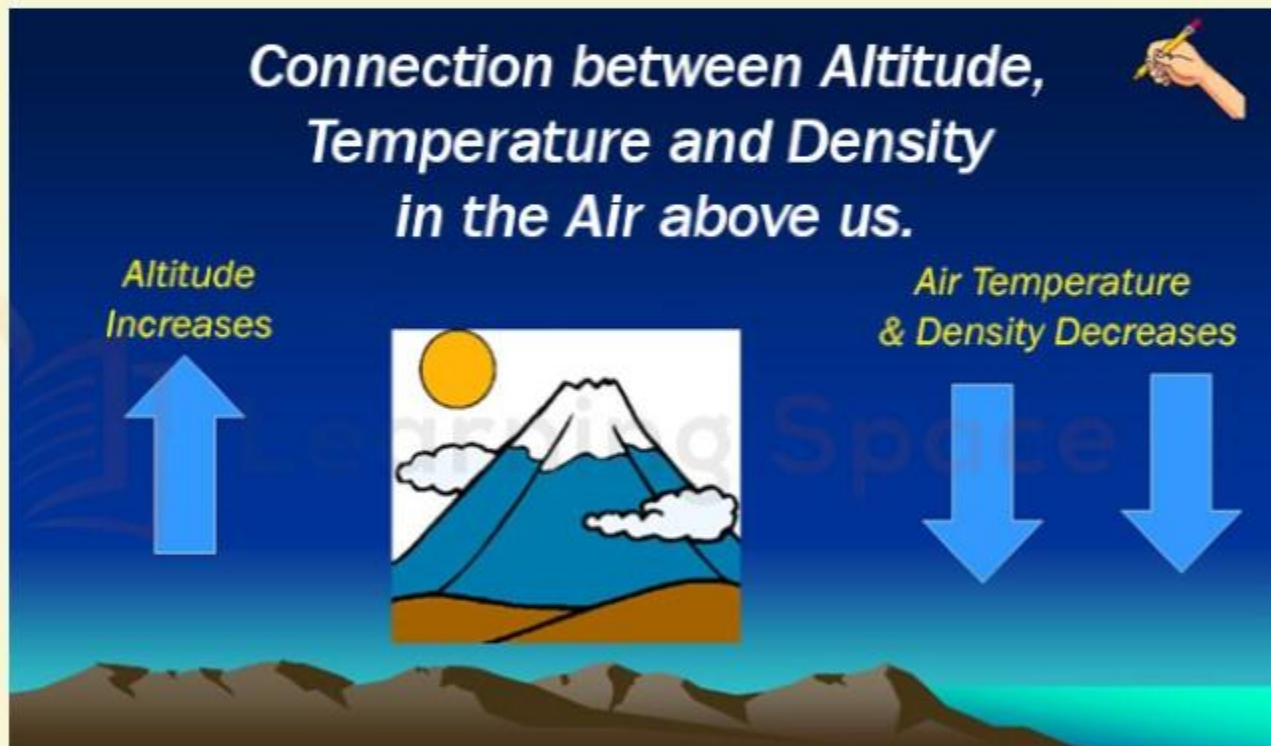


Why does this temperature drop happen?

- 1** As we ascend higher in the atmosphere, the **amount of heat gets lost to subsecutive layer** lying below.
- 2** The air pressure is higher in the lower portion of the atmosphere near the earth's surface because of the weight of all the overlying layers above. Thus **density of air decreases** as we move upwards.
- 3** The **lower layer of the air contains more water vapour and dust particles** than the layers above and hence it **absorbs larger amount of heat radiated** from the earth's surface than the upper layers.

STRUCTURE OF THE ATMOSPHERE

1.TROPOSPHERE



STRUCTURE OF THE ATMOSPHERE

1.TROPOSPHERE

Normally, the temperature decreases with the increase in height from the Earth's surface, because

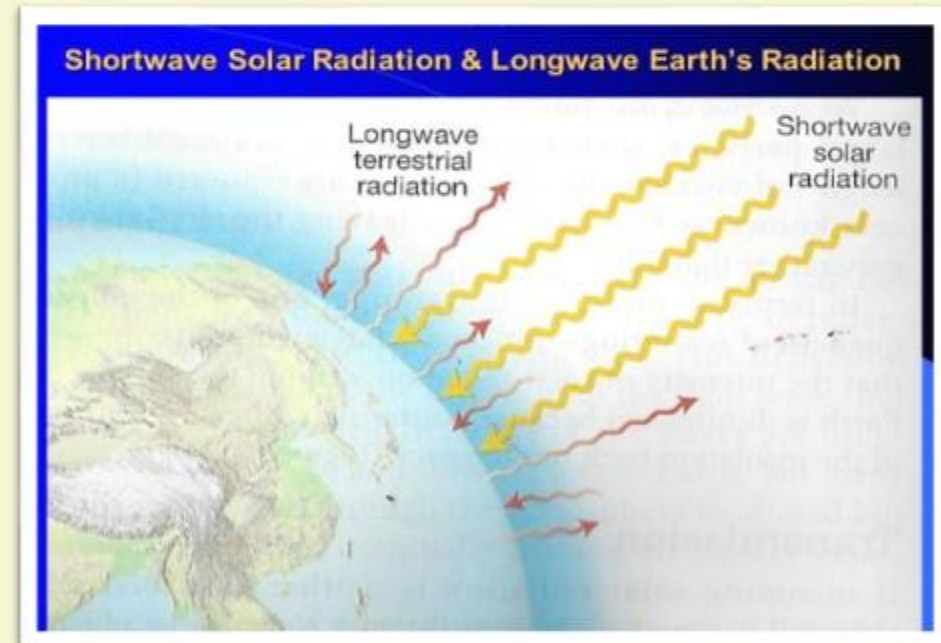
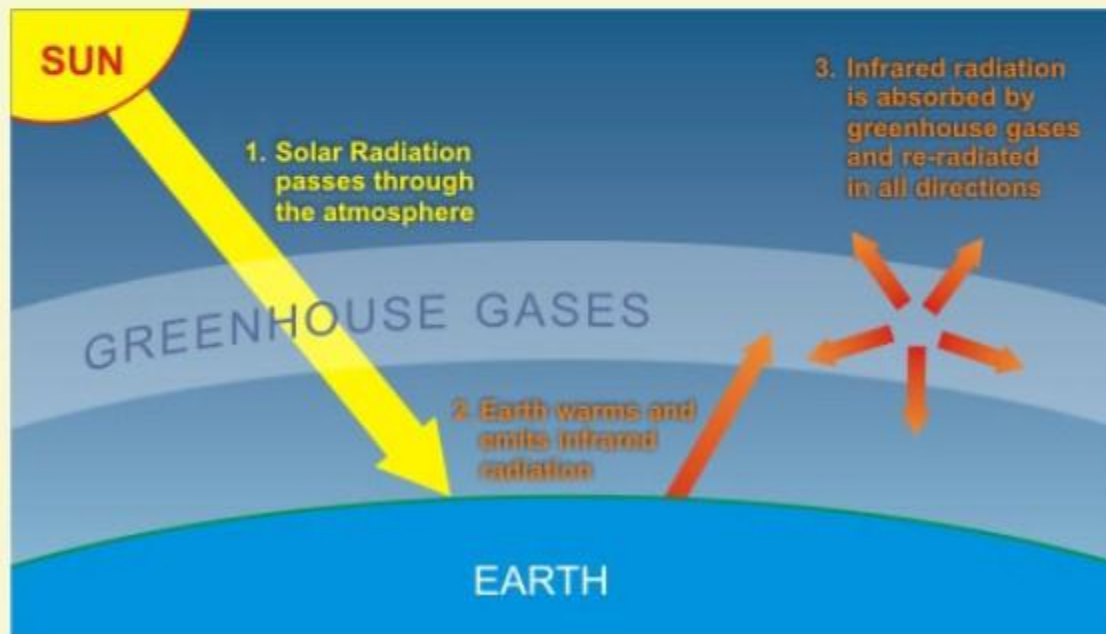
1. The atmosphere can be heated upwards only from the Earth's surface
2. There is more moisture in the upper atmosphere
3. The air is less dense in the upper atmosphere **(UPSC Prelims 2012)**

Select the correct answer using the codes given below :

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only**
- d) 1, 2 and 3

EXPLANATION

- Statement 1 is correct since atmosphere is heated by infrared radiation emitted by earth. That is, **long-wave terrestrial radiation rises the temperature** of the earth than incoming short-wave radiation. This source of heat is known as **ground-radiation or terrestrial radiation**.
- 2nd statement is not true. In fact, it is the opposite. More than 90% of water vapour or moisture present within 5km from earth surface.
- 3rd statement is true. Air is less dense; meaning it can hold less heat. Thus temperature is low in the upper atmosphere.



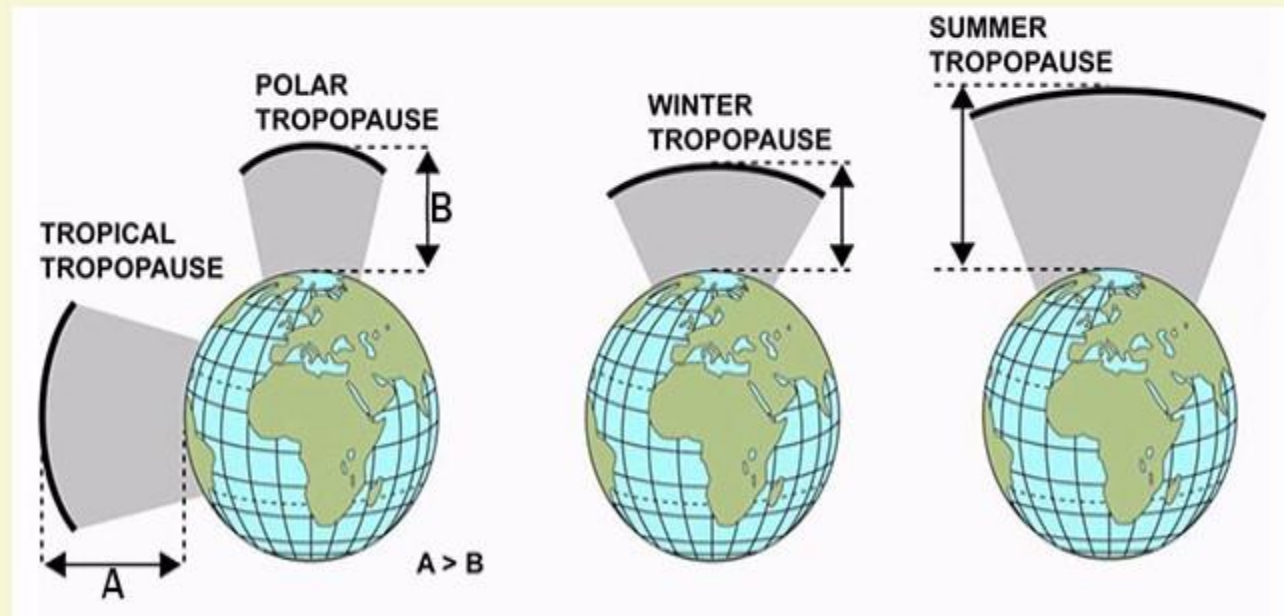
STRUCTURE OF THE ATMOSPHERE

1.TROPOSPHERE

- 6 There is a seasonal difference of height i.e. it varies by season. It is lower in winter and higher in summer.

What is the reason for seasonal variation of height?

- 1 In winters, the air becomes cold i.e. becomes more dense. Therefore in general there is a drop in height.
- 2 Over the equator, there is no winter as there is less seasonal differences. Hence this variation is not greatly observed.
- 3 However, as we move towards the poles from the equator, the height increases during summer and decreases during winter.



STRUCTURE OF THE ATMOSPHERE

1.TROPOSPHERE

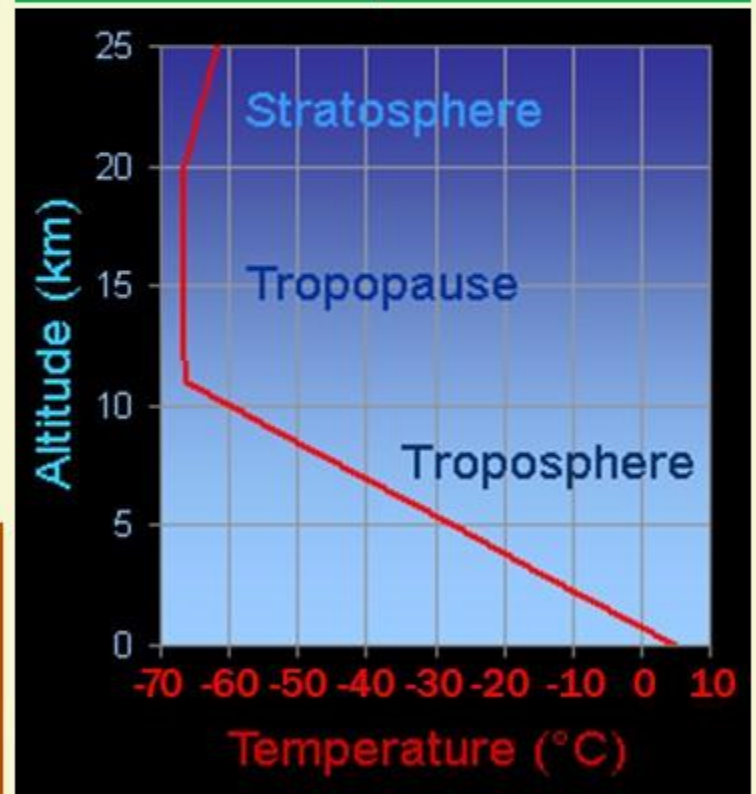
- 7** There is an altitudinal variation of temperature in troposphere.

What happens at the tropopause?

- 1** The zone separating the troposphere from stratosphere is known as the tropopause.
- 2** Tropopause is about 1.5 km thick
- 3** This is an isothermal layer as temperature gradient here is nearly constant
- 4** The air temperature at the tropopause is about minus 80°C over the equator and about minus 50°C over the poles.

Beware temp is lowest above equator and NOT the other way round!!

Tropopause is an isothermal layer

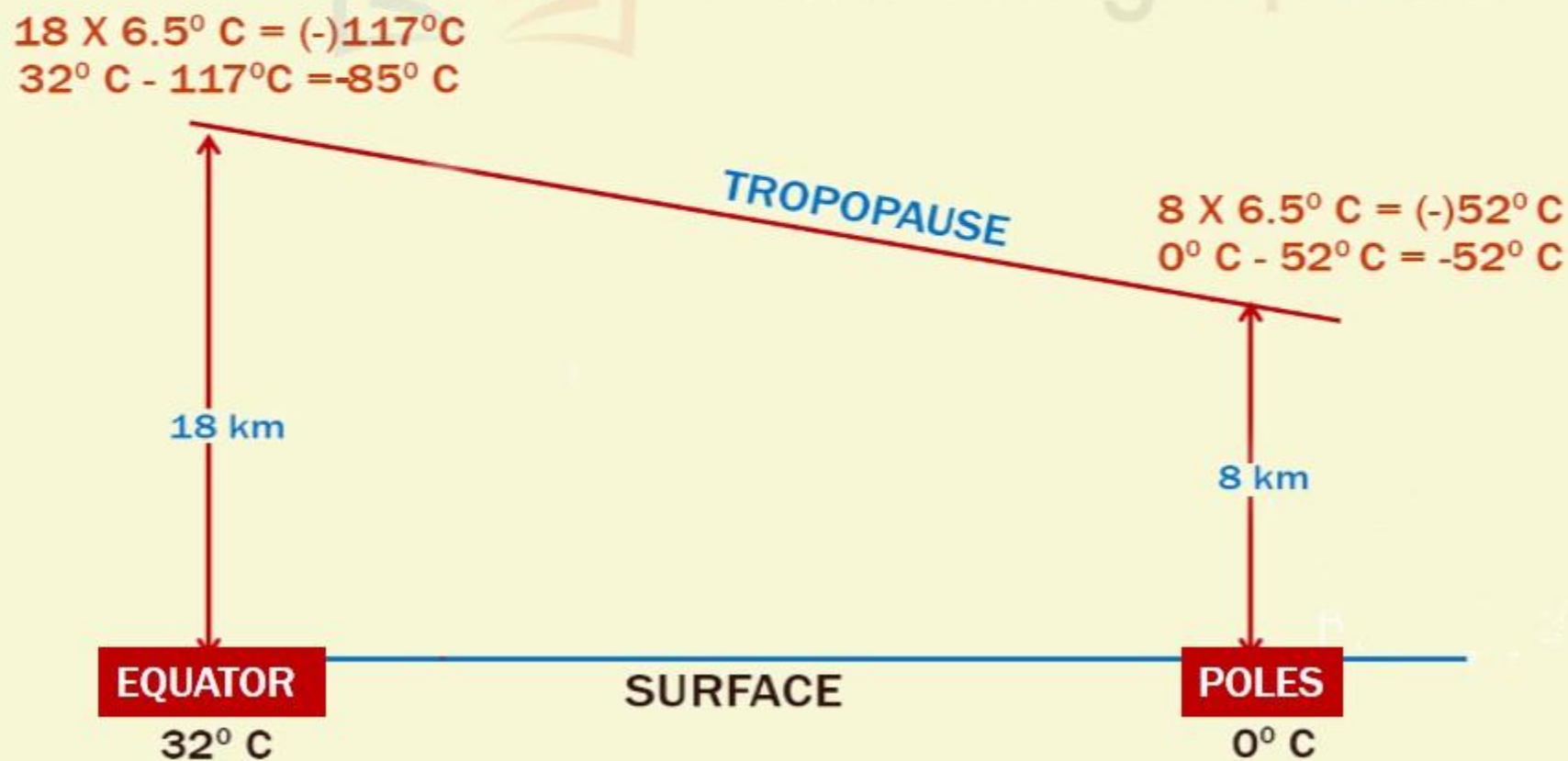


STRUCTURE OF THE ATMOSPHERE

1.TROPOSPHERE

Why is temperature lowest at the tropopause above the equator than at the poles?

Since temperature decreases upward at the rate of 6.5°C per 1000m and hence it is natural that at the height of 18 km over the equator becomes much lower than at the height of 8 km over the poles.



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